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Eugene Shakhnovich and Alexander Gutin

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Enumeration of all compact conformations of copolymers with random sequence of links

Eugene Shakhnovich

Department of Chemistry, Harvard University, Cambridge, Massachusetts 02138

Alexander Gutin

Institute of Protein Research Ac. Sci. of the USSR, 142292 Puschino Moscow Region, USSR

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Exhaustive enumeration of all compact self-avoiding conformations of a chain of 27 monomers on the $3 \times 3 \times 3$ fragment of a simple cubic lattice is given. Total number of conformations unrelated by symmetry is 103 346. This number is relatively small which makes it possible to make a numerically exact calculation of all thermodynamic functions this chain.

Heteropolymers with random sequence of links were considered, and the freezing transition at finite temperature was observed. This transition is analogous to folding transition in proteins where unique structure is formed. The numeric results demonstrate the equivalence between random 3-dimensional heteropolymers and the random energy model found previously in analytical investigations. The possible application of these results to some problems of combinational optimization is discussed.

I. INTRODUCTION

Configurational statistics of random heteropolymers had been intensively studied during last few years.¹⁻⁵ The most remarkable feature of heteropolymeric chains is the existence of a freezing transition at finite temperature T_c . Configurational entropy per monomer vanishes below T_c . This means that at $T < T_c$ only few (~ 1) conformations with low energy dominate. This is like folding transition in proteins where one conformation (native) dominates for a given sequence.

Analytical investigation of random copolymers³⁻⁵ demonstrates deep analogy between heteropolymeric chains and models of spin glasses. It was assumed, e.g., in Ref. 3 and shown in Ref. 5 that heteropolymers in 3-dimensional space can be described by the random energy model (REM)⁶ introduced by Derrida in the theory of spin glasses. In fact, the freezing transition in heteropolymers is analogous to the glass transition in the REM.

It was argued in Ref. 5 that overall compactization of a chain (increase of its density) facilitates the freezing transition (it increases the transition temperature T_c). This leads to the idea of numerically exact procedure to test these results. The reason is that it is possible to scan not all tremendously large number of conformations of a chain but only compact ones—Hamiltonian paths on some lattice. Another reason to consider only compact conformations of a chain is that the most natural application of these results may be for investigation of thermodynamics of protein folding. One should take into account, however, that protein globules are very compact with density in the interior close to such of molecular crystals.⁷

Consideration of only compact conformations provides a great technical advantage since it decreases the number of conformations significantly. Indeed, the number of all conformations of a chain of N monomers on a lattice with coordination number z is z^N while the number of compact conformations is of the order $(z/e)^N$ (see, e.g., Refs. 8 and 9).

This makes it possible to make exhaustive enumeration of all compact self-avoiding walks of relatively long chains and hence to obtain numerically exact description of random heteropolymer.

II. MODEL AND ALGORITHM OF ENUMERATION

A freely joint chain with unit bond length composed of 27 monomers was placed in a fragment of a cubic lattice of the size $3 \times 3 \times 3$. Self-avoiding walks of this chain were considered (Fig. 1). A first-depth algorithm was applied to make an exhaustive enumeration of conformations.

First, we introduce connectivity of the lattice—matrix $C(m,n) = 1$ if sites m and n are neighbors in the lattice and $C(m,n) = 0$ otherwise. Next, we position nonzero elements of C in some (arbitrary) order and hence obtain for any site m the set of “directions” at which one can get from this site to nearest-neighbor sites. Then, starting from some initial position of the first monomer of the chain we grow it step by step positioning subsequent monomers in directions allowed by matrix C . In this case the result of i steps—conformation of i -monomer part of the chain—is determined by the position of the first monomer m_1 and the set of directions $a_1 a_2 \dots a_{i-1}$ which can be easily transferred to the set of positions $m_1 m_2 \dots m_i$. When we come to a dead end we make one step back and try to move in the next direction, i.e., try a conformation $\{m_1; a_1 a_2 \dots (a_{i-1} + 1)\}$, etc. This procedure is repeated until successful termination is reached when all 27 monomers are positioned in the lattice. Then in order to search for a new conformation we make a step back and try the next direction of the last link; then all procedure is repeated. The essential feature of this algorithm is that we tract successful terminations in the same manner as dead ends with the only difference that successful terminations are memorized. It makes it possible to try *all* available directions a_i and, hence, to search conformations exhaustively.

There are a lot of conformations which are related by symmetry. Some of them are related by rigid body rotations

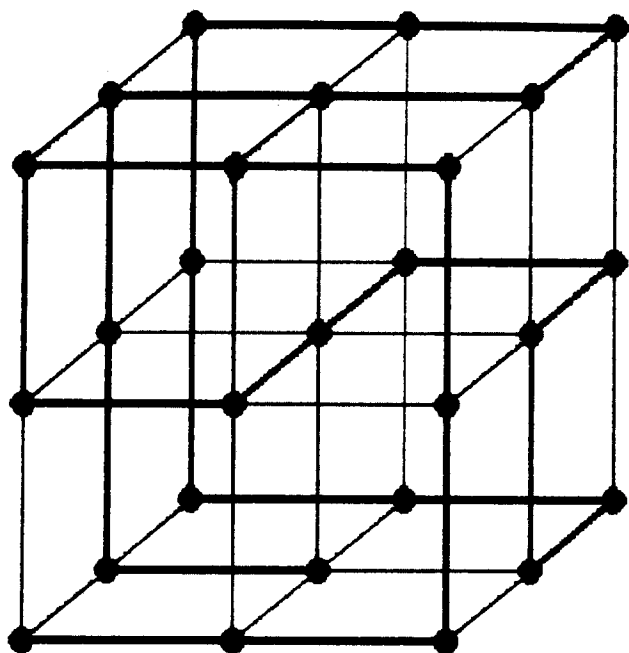


FIG. 1. The fragment lattice and example of a compact self-avoiding walk of a chain of 27 monomers (fat line).

of the cube, some of them are related by mirror symmetry. Breaking of this symmetry can be achieved by positioning of a few first monomers "by hands." The problem of choice of initial conditions is simplified by a following simple *lemma*.

Hamiltonian paths on $3 \times 3 \times 3$ fragment of cubic lattice can start only from vortexes or centers of facets of the cube.

The proof of this lemma was given by Razgulyaev (O. I. Razgulyaev, personal communication). Divide lattice sites into two classes: to the first (class *A*) belong vortexes and centers of facets, to the second (class *B*) all other sites. It is easy to see that there are $N_A = 14$ elements in class *A* and $N_B = 13$ elements in class *B*. Any conformation can be presented in these terms as *ABABABA...* since position of a subsequent monomer must belong to the opposite class than position of the previous one (this is due to the fact that the length of a bond is 1). The results follows from the fact that $N_A - N_B = 1$ and hence any conformation must start and end in a site of *A*-class, i.e., in a vortex or a center of a facet.

As a result all compact self-avoiding walks unrelated by symmetry can be found for this model. The total number of conformations is 103 346. This number is relatively small to calculate all thermodynamic functions of a chain.

III. RESULTS

Energy E_k is associated with each conformation k :

$$E_k = \frac{1}{2} \sum_{i,j}^N B_{ij} C(m_i, m_j) \quad (1)$$

summation in (1) is taken over all links of the chain which are not nearest neighbors along the chain (interactions between them give a constant contribution to energy and thus unessential). The presence of connectivity matrix in (1) is

due to the fact that only interactions between nearest neighbors in space are taken into account. Interaction energies B_{ij} characterize sequence and are taken as random values with Gaussian distribution:

$$P(B_{ij}) = \frac{1}{(2\pi B^2)^{1/2}} \exp\left(-\frac{B_{ij}^2}{2B^2}\right), \quad (2)$$

where B is standard variance. This choice of probability density (PD) corresponds to heteropolymers with large number of types of links; the same PD for B_{ij} was assumed in analytical investigations.³⁻⁵

Different "sequences" are different sets of B_{ij} generated according to PD (2) with $B = 1$. We generated 200 "sequences" and for each of them calculated partition functions and Boltzmann probabilities of each conformation:

$$Z_l = \sum_{k=1}^{103\,346} \exp\left(-\frac{E_k^l}{k_B T}\right)$$

and

$$p_k^l = \exp\left(-\frac{E_k^l}{k_B T}\right) / Z_l, \quad (3)$$

here index l ($l = 1, 2, \dots, 200$) denotes sequence and index k denotes conformation. k_B is Boltzmann constant and T is temperature.

First of all we determined how many conformations are thermodynamically relevant at a given temperature, i.e., investigated freezing transition. **The following value characterizes the number of thermodynamically relevant conformations.**^{10,11}

$$X_l = 1 - \sum_k (p_k^l)^2, \quad (4)$$

where $M = 103\,346$ is the total number of conformations. Indeed, at high temperature when all conformations have equal Boltzmann probabilities $p_k^l = 1/M$ then $X_l = 1 - 1/M \approx 1$. In the opposite case when the chain is completely frozen so that only one conformation $k = k_0$ dominates and $p_{k_0}^l = 1$ while $p_k^l = 0$ for $k \neq k_0$ we have $X_l(T) = 0$.

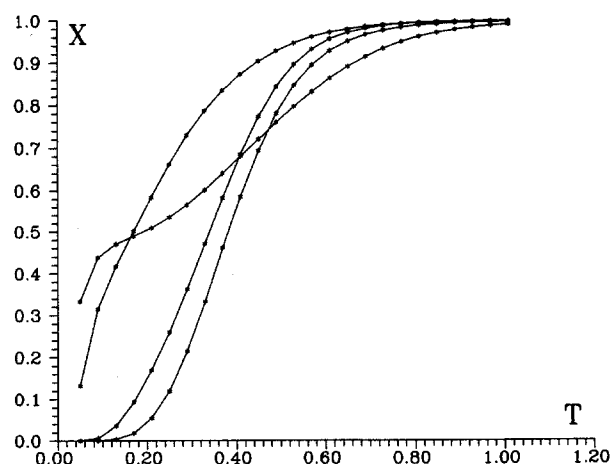


FIG. 2. Plot of $X_l(T)$ for some sequences.

The results of calculations for some sequences are shown in Fig. 2. One can see that each sequence undergoes a freezing transition at $T \approx T_c$. For $T > T_c$ $X_i \equiv 1$ while for $T < T_c$ $X_i < 1$ and depends on temperature T . This transition is rather smooth due to finiteness of a chain. It is interesting to note that curves $X_i(T)$ for different sequences converge in the vicinity of the transition point T_c and diverge at lower temperatures. **This means that the transition temperature is the same for any sequence, i.e., it is a self-averaging value.** However, X_i is not self-averaging at $T < T_c$: it fluctuates strongly from sequence to sequence. This is due to the fact that in the frozen state X_i is determined by few states with lowest energies and depends on particular features of the bottom part of the spectrum of energies of conformations.

The convergence of curves in Fig. 2 when $T \rightarrow T_c$ implies that the histogram for probability distribution of $X(T)$ becomes narrow in this temperature region and it is broad at lower temperatures. Histograms shown in Fig. 3 confirm this conclusion. It is interesting to note that the histogram in Fig. 3(b) is bimodal. This confirms a very delicate result of Ref. 10, where this bimodality has been obtained analytically.

We examined also $\langle X(T) \rangle = (\sum_i^{200} X_i)/200$ which is the result of averaging over all sequences. (Everywhere below $\langle \rangle$ will denote averaging over all sequences.) According to Ref. 5 $\langle X(T) \rangle = T/T_c$ for $T < T_c$ and $\langle X(T) \rangle = 1$ for $T > T_c$. The numeric result is shown in Fig. 4. It can be

seen that there is indeed a straight part of the curve at low temperature.

The very important feature of the frozen state of heteropolymeric chains is that for a given sequence the states with lowest energies (and only these states are relevant in the frozen phase) are structurally different.⁵ This makes ground to treat energies of these states as independent random values in the same manner as it was assumed in the REM. We tested this feature in the numeric model. **Similarity overlap** between two conformations, say, k and k' , is defined as

$$Q_{kk'} = \frac{N_{\text{common}}(k, k')}{N_{\text{total}}}, \quad (5)$$

where $N_{\text{common}}(k, k')$ is the number of contacts which are the same in conformations k and k' —contacts between the same monomers i and j . N_{total} is the total number of long-range contacts in each conformation. For the lattice fragment which we consider $N_{\text{total}} = 28$.

The total number of conformations M is so large that we may use probabilistic description of their mutual overlaps. Consider probability $P_l(Q)$ that **two conformations for a given sequence l taken randomly but with corresponding Boltzmann probabilities will have overlap Q .** The formal definition is

$$P_l(Q) = \sum_{k, k'}^M \delta(Q - Q_{kk'}) p_k^l p_{k'}^l. \quad (6)$$

Introduction of Boltzmann probabilities p_k^l into definition of $P_l(Q)$ means that in the frozen phase only few conformations with lowest energies give contribution to $P_l(Q)$ and, hence, structural similarity (or difference) of these ground states is reflected by $P_l(Q)$.

To make definition of $P_l(Q)$ more clear it is reasonable to split the sum in Eq. (6) into two parts: terms with $k = k'$, i.e., self-overlaps (of course, $Q_{kk} = 1$) and terms with $k \neq k'$ which correspond to overlaps between different conformations. Using Eq. (4) we obtain

$$P_l(Q) = (1 - X_l) \delta(Q - 1) + \sum_{k \neq k'}^M \delta(Q - Q_{kk'}) p_k^l p_{k'}^l. \quad (7)$$

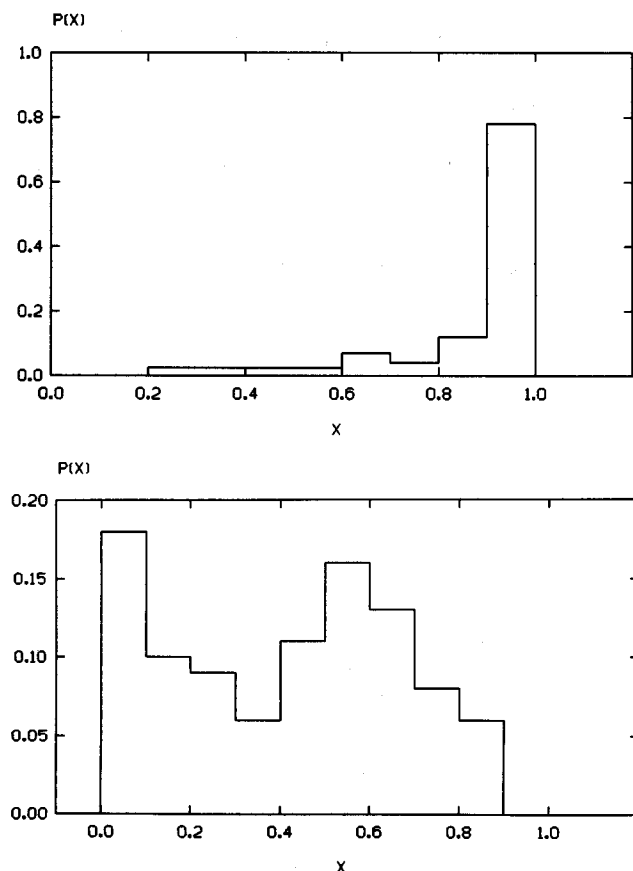


FIG. 3. Histograms of X for 200 sequences. (a) Histograms for $T = 0.9T_c$. (b) Histograms for $T = 0.5T_c$.

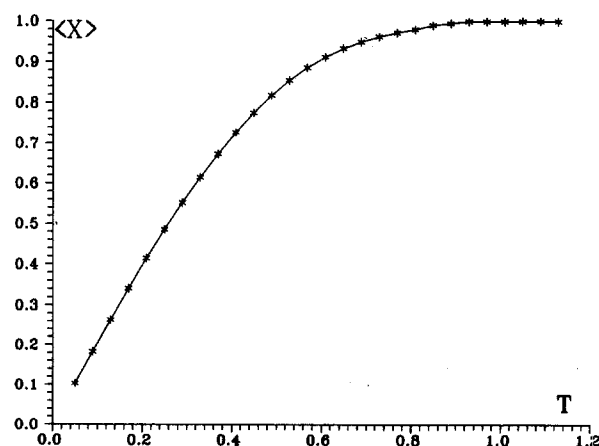


FIG. 4. Plot of $\langle X(T) \rangle$.

We see that $P_i(Q)$ must have peak of height $1 - X_i$ at $Q = 1$. However, the behavior of $P_i(Q)$ at $Q < 1$ characterizes how similar are different conformations with low energies.

We calculated $\langle P_i(Q) \rangle$ as a result of averaging of individual $P_i(Q)$ over all 200 realizations of sequences. The result is shown in Fig. 5(a). One can see that the histogram in Fig. 5(a) is pronouncely bimodal, so that conformations contributing to this histogram are not structurally close to each other. However, the peak concentrated at small overlaps Q is smooth and shifted rightwards. We suggest that this is due to two reasons: (1) chain connectivity (so that contacts between monomers which are close to each other in sequence are more probable than contacts between monomers which are distant in the sequence) and (2) finiteness. In order to check it we took randomly 50 conformations from the list and assumed equal probability $p_k = 0.02$ for each conformation [Fig. 5(b)]. Comparison of Fig. 5(a) and Fig. 5(b) shows that $\langle P(Q) \rangle$ at $Q < 1$ is hardly distinguishable from overlaps of randomly chosen conformations and hence there is no essential structural similarity between different conformations with low energies.

Evaluation of $\langle P(Q) \rangle$ makes it possible to find a Parisi order parameter $Q(X)$ for heteropolymeric chain.⁵ Using the identity¹³

$$\frac{dX}{dQ} = \langle P(Q) \rangle \quad (8)$$

it is possible to evaluate numerically $Q(X)$. The result is shown in Fig. 6. One can see that $Q(X)$ is close to step function, and position of the breakpoint tends to 0 when $T \rightarrow 0$. However $Q(X) > 0$ when $X \rightarrow 0$. This indicates the fact that there are always some structural overlaps between conformations of a chain.

IV. DISCUSSION

The presented results give some insight in features of random heteropolymers of moderate size. It should be emphasized that these results confirm even the most delicate predictions of the analytical theory of heteropolymers³⁻⁵ and the REM.⁶ This is not trivial since asymptotics of large chain length is always assumed in statistical mechanics calculations. We see that this asymptotics is reached even at $N = 27$. This is very important for application of the theory to investigation of protein folding.

Two conclusions seem to be most important. First, is the small structural similarity (almost as in randomly chosen conformations, see Fig. 5). This means that thermal excitations in the system which cause transitions from the ground state to the closest to it "excited" state demand significant rearrangement of the whole structure instead of small shifts of few monomers.

Second conclusion is the observation of the absence of self-averageness of $X_i(T)$. We would like to stress once again that even bimodality of distribution of $X_i(T)$ predicted in Ref. 10 is observed. This may be important for understanding of evolution of protein structures. Considering creation of stable proteins, e.g., which have pronounced ground fold dominance as the aim of evolution on the structural level we can see that this aim can be reached at relatively low price: a significant fraction of random sequences will have $X_i \ll 1$ at $T < T_c$ (see the more detailed discussion of this approach in Ref. 12).

The above results may have some general methodic importance. This is one of the examples when "spin-glass like"

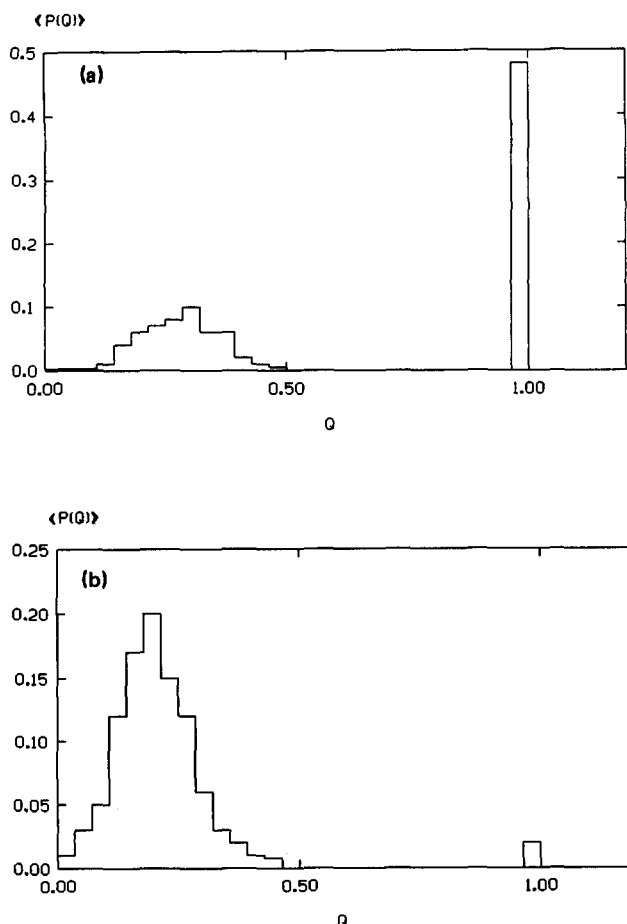


FIG. 5. Histograms of overlap between different configurations. (a) Histogram for $\langle P(Q) \rangle$. (b) Histogram for mutual overlaps between 50 structures taken randomly with equal weights $p_k = 0.02$.

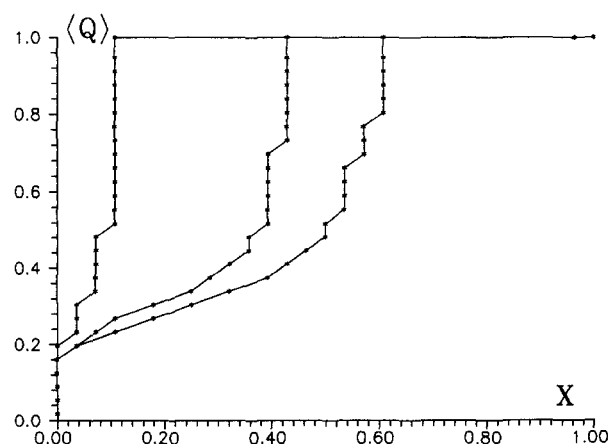


FIG. 6. Plot of Parisi order parameter $\langle Q(X) \rangle$ at different temperatures: $T_1 = 0.1T_c$, $T_2 = 0.2T_c$, $T_3 = 0.5T_c$. The position of the step moves rightwards when temperature increases.

behavior is observed as a result of exhaustive enumeration of all microconfigurations. It is possible also to investigate some other models of disordered chains, e.g., the case of "two-letter" heteropolymer (random sequences of polar and unpolar monomers) which are difficult to analyze analytically.

Some interesting questions were not addressed here. First of all, it is interesting to investigate effect of size. This is not possible to do in $3D$ lattices since there are only three unrelated by symmetry walks on fragment $2*2*2$ of a cubic lattice and this number is too small to draw any statistical conclusions. Calculations on the fragment $4*4*4$ are out of reach now due to tremendously large number of compact conformations. However, it is possible to make exhaustive search of all conformations on square lattices of different sizes. This was done in the recent work¹³ for the case of two-letter random heteropolymers.

Exhaustive enumeration of microconfigurations by first-depth algorithm may be useful in some problems of combinatorial optimization.¹⁴ Considering a TSP problem on a graph with finite connectivity, e.g., we may draw an

analogy between routes of a salesman as flights of a chain. Then the procedure analogous to the one described above will give exhaustive enumeration of all routes.

This may enhance our intuition which is a base point for construction of any analytical theory.

¹A. Yu. Grosberg and E. I. Shakhnovich, *Soviet Phys. (JETP)* **64**, 1284 (1986).

²T. Garel and H. Orland, *Europhys. Lett.* **6**, 307 (1988).

³J. B. Bryngelson and P. G. Wolynes, *Proc. Natl. Acad. Sci. USA* **84**, 7524 (1987).

⁴E. I. Shakhnovich and A. M. Gutin, *J. Phys. A* **22**, 1647 (1989).

⁵E. I. Shakhnovich and A. M. Gutin, *Biophys. Chem.* **34**, 187 (1989).

⁶B. Derrida, *Phys. Rev. B* **24**, 2613 (1981).

⁷E. I. Shakhnovich and A. V. Finkelstein, *Biopolymers* **28**, 1667 (1989).

⁸H. Orland, C. Itzykson, and C. De Dominicis, *J. Phys. (France) Lett.* **46**, L-353 (1985).

⁹K. Dill, *Biochemistry* **24**, 1501 (1985).

¹⁰M. Mezard, G. Parisi, N. Sourlas, G. Toulouse, and M. Virasoro, *J. Physique (France)* **45**, 843 (1984).

¹¹B. Derrida and G. Toulouse, *J. Physique (France) Lett.* **46**, L223 (1985).

¹²E. I. Shakhnovich and A. M. Gutin, *Nature* **346**, 773 (1990).

¹³K. F. Lau and K. A. Dill, *Macromolecules* **22**, 3986 (1989).

¹⁴M. Mezard, G. Parisi, and M. Virasoro, *Spin Glass Theory and Beyond* (World Scientific, Singapore, 1988).